Acid–Base Balance by the Kidneys

Undergraduate medical physiology

Urogenital and endocrine module

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Learning objectives

Explain pH and the bicarbonate–CO_2 relationship.

Trace bicarbonate reclamation along the nephron.

Compare reclamation with net new bicarbonate generation.

Predict renal responses to acidosis and alkalosis.

Interpret primary disorders and expected compensation.

Follow the movement of an acid or bicarbonate equivalent.

Do you remember ?

- All the cells undergo a metabolism produce ATP-energy from proteins, fats and carbohydrates.
- Proteins and carbohydrates are made of carbons, Hydrogen's and oxygen's.(C, H, O2)
- During metabolism process: one hydrogen is removed and transported to the mitochondria to undergo a process called electron transport chain for making ATP.
- The remained Carbon and oxygen together are carbon dioxide (CO2).
- This C02 must be removed from our body.

- To remove CO2 from our body to the lungs, firstly, it reacts with the water in blood stream to form an acid (Carbonic acid).
- The carbonic acid is self dissociative and forms H+ and bicarbonate ions.

$$CO_2 + H_2O \longrightarrow H_2CO_3 \rightleftharpoons H^+ + HCO_3$$

CO2 removal is handled by our lungs during exhalation.

Bicarbonate and H ion removal is handled by our kidneys.

Acid and Base definitions

- A hydrogen ion is a single free proton released from a hydrogen atom.
- H+ shows a very strict effect on physiological activities.
- Molecules containing hydrogen atoms that can release hydrogen ions in a solution are called acids.
- An example is hydrochloric acid (HCl), which ionizes in water to form hydrogen ions (H+) and chloride ions (Cl−).
- Carbonic acid (H2CO3) ionizes in water to form H+ and HCO3−.

- A base is an ion or a molecule that can accept an H+.
- For example, HCO3− is a base because it can com bine with H+ to form H2CO3.
- HPO42− is a base because it can accept an H+ to form H2PO4−.
- The proteins in the body also function as bases because some of the amino acids that make up proteins have net negative charges that readily accept H+.
- The protein hemoglobin in the red blood cells and proteins in other cells of the body are among the most important of the body’s bases.

Acids, bases, and pH

Acids donate H^+. Bases accept H^+.

Strength describes dissociation, not concentration.

Acidosis / alkalosis describe processes.

Acidemia <7.35. Alkalemia >7.45.

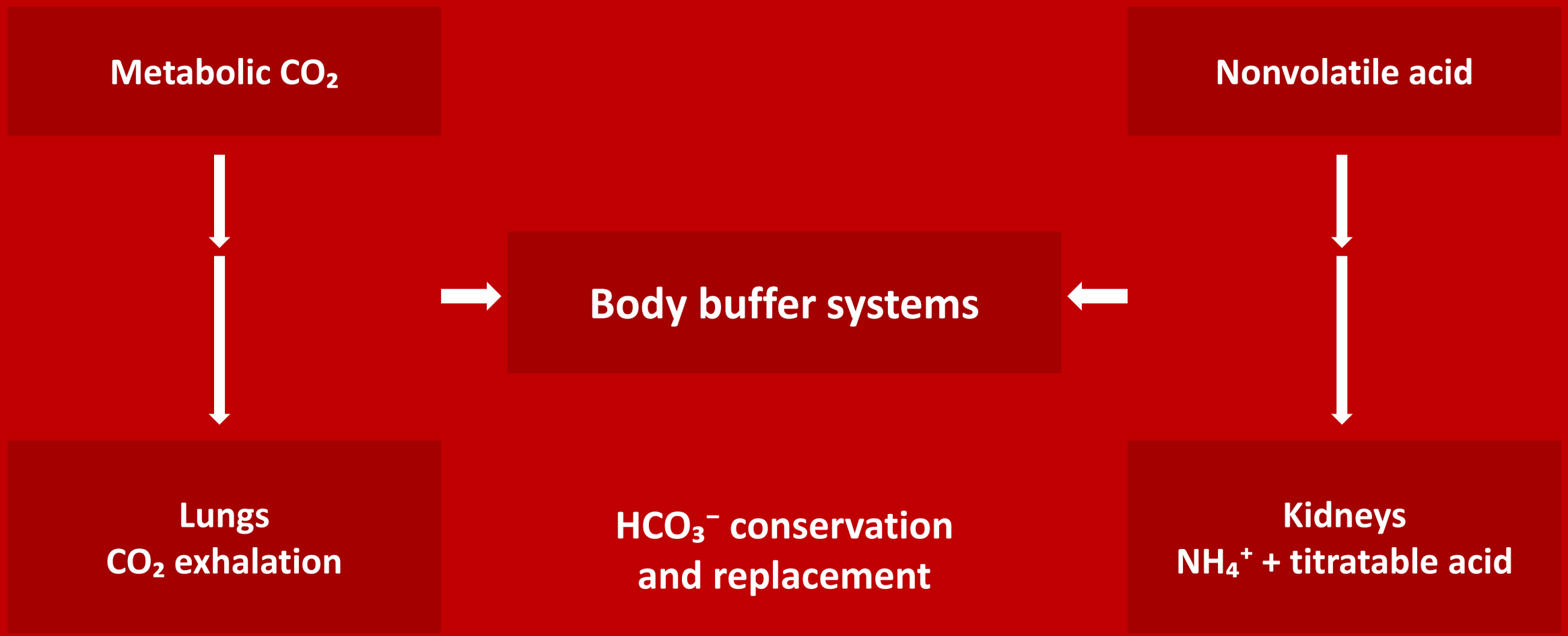
$$pH = -\log_{10}[H^+]$$

pH	$[H^+]$, nmol/L
7.30	≈50
7.40	≈40
7.50	≈32

$[H^+]$ in mol/L for the logarithm. Normal arterial pH: 7.35–7.45.

A fall of one pH unit means a tenfold rise in $[H^+]$.

Metabolic acid and whole-body balance



Arrows show the route of acid handling. Buffers limit the initial pH change.

The lungs remove CO_2. The kidneys handle net nonvolatile acid.

The bicarbonate buffer system

$$CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$$

Carbonic anhydrase

Added H^+ consumes bicarbonate and promotes CO_2 formation.

The lungs remove CO_2 while the kidneys regulate bicarbonate.

Proteins and phosphate also buffer acid in their compartments.

Bicarbonate buffering works with an open route for CO_2 elimination.

The Henderson–Hasselbalch relationship

$$pH = 6.1 + \log_{10} \frac{[HCO_3^-]}{0.03 \times PCO_2}$$

Term	Meaning / clinical units
6.1	Apparent pK near 37°C
$[HCO_3^-]$	Bicarbonate concentration, mmol/L
PCO_2	CO_2 partial pressure, mmHg
0.03	CO_2 solubility, $mmol \cdot L^{-1} \cdot mmHg^{-1}$

24 / (0.03 × 40) = 20 pH = 6.1 + $\log_{10}$20 ≈ 7.40

pH depends on the bicarbonate-to-dissolved-CO_2 ratio.

Three defences against pH change

Defence	Approximate time	Physiological action
Chemical buffers	Seconds	Bind or release H^+
Lungs	Minutes	Adjust CO_2 elimination
Kidneys	Hours to days	Reclaim HCO_3^- and adjust acid excretion

These are overlapping responses, not durations after which a defence stops.

Retrieval: which defence removes nonvolatile acid?

These responses overlap in time and act together.

Renal bicarbonate and acid handling

4,320

mmol of HCO_3^- filtered per day

180 L/day × 24 mmol/L

Illustrative GFR × plasma concentration

1-Reclaim filtered bicarbonate.

2-Replace bicarbonate consumed by net acid.

3-Excrete acid with urinary buffers.

GFR = glomerular filtration rate. For monovalent HCO_3^-, 1 mmol = 1 mEq.

Reclamation protects the existing pool. Net acid excretion replenishes it.

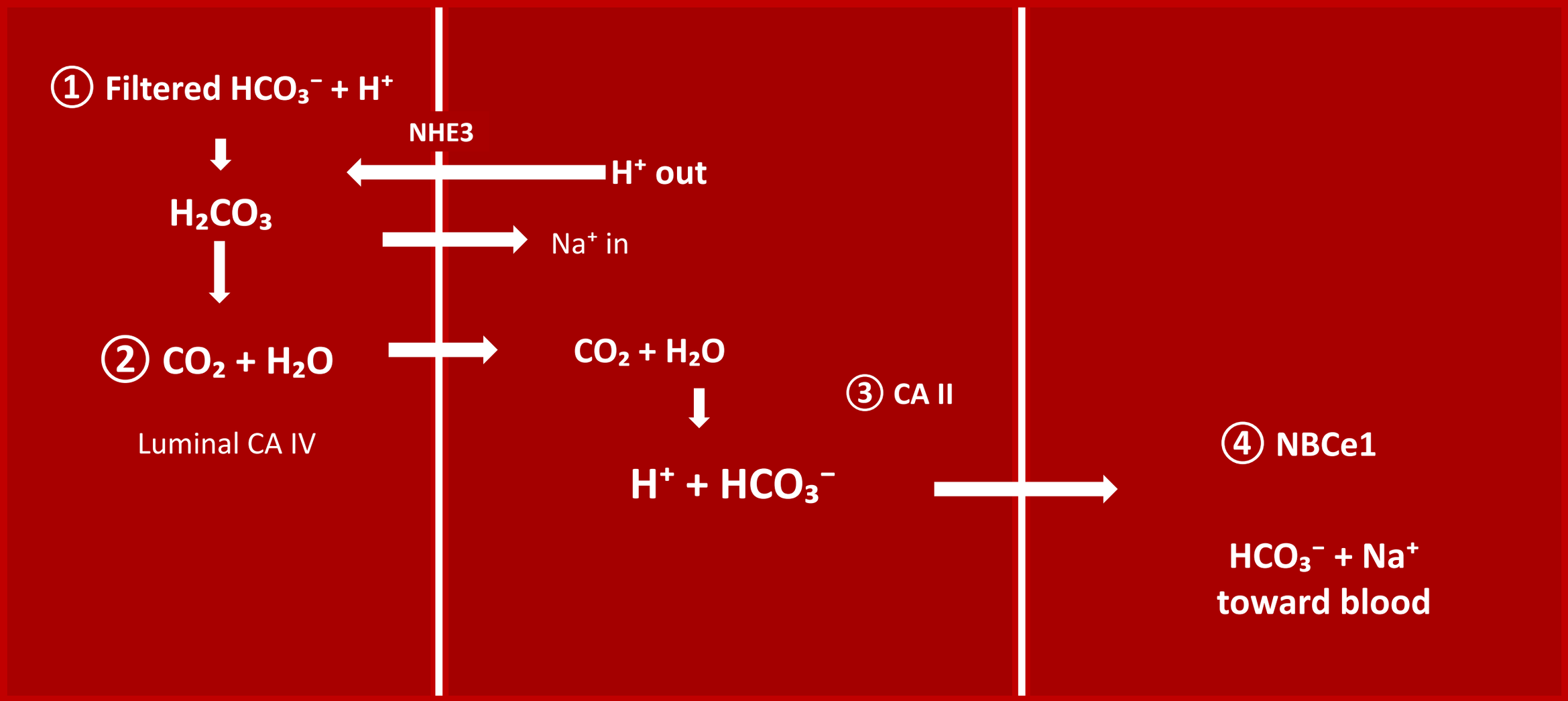
Sites of renal acid–base transport

Glomerular filtrate	→	Proximal tubule	→	Thick ascending limb	→	Late distal nephron / CD
Filtered HCO_3^-		≈80–90% reclaimed NHE3; ammonia production		≈10–15% reclaimed NH_4^+ recycling		Remainder reclaimed Final H^+ secretion

Functional map, not anatomical scale. Approximate fractions vary. CD = collecting duct.

Different nephron segments cooperate in the final urine balance.

Proximal bicarbonate reclamation



CA = carbonic anhydrase. H^+ is recycled. Na^+/K^+-ATPase maintains the Na^+ gradient.

H^+ is recycled while a filtered bicarbonate equivalent returns to blood.

Reclaimed and new bicarbonate

RECLAMATION

H^+ + filtered HCO_3^-

↓

Filtered equivalent returns to blood

Net new HCO_3^-: zero

NEW BICARBONATE

H^+ + nonbicarbonate buffer

↓

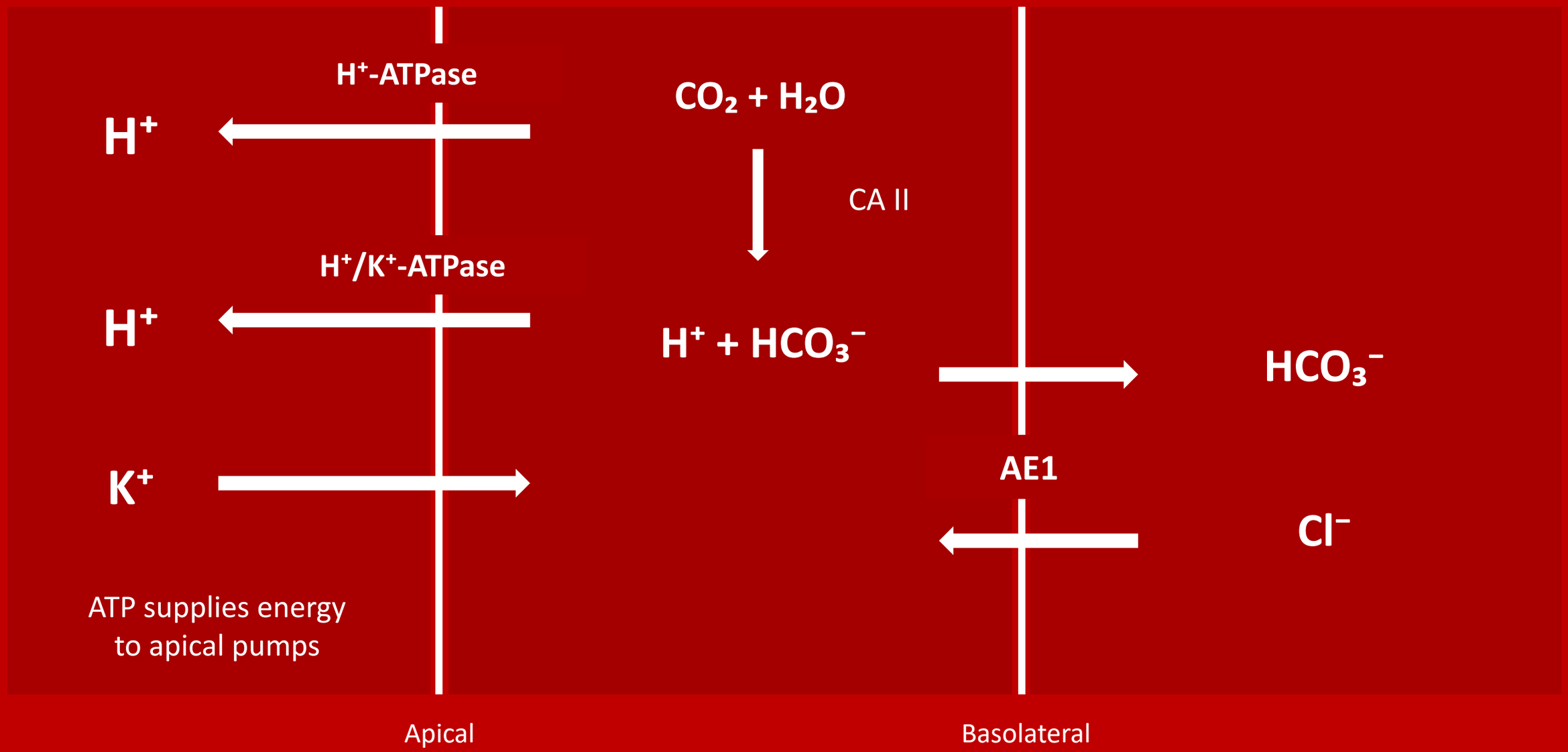
H^+ leaves in urine
HCO_3^- enters blood

Net new HCO_3^-: one equivalent

For ammonium, urinary excretion must follow renal production to yield a net systemic gain.

The fate of secreted H^+ determines the net bicarbonate effect.

The alpha-intercalated cell



Arrowheads show net ion movement. CA = carbonic anhydrase. AE1 = anion exchanger 1.

Membrane polarity directs acid toward urine and bicarbonate toward blood.

Phosphate buffers secreted H^+

$$HPO_4^{2-} + H^+ \rightleftharpoons H_2PO_4^-$$

↓

Titratable acid in urine

Free H^+ at urine pH 4.5

0.0316 mmol/L × 1.5 L/day ≈ 0.047 mmol/day

Illustrative calculation. Phosphate is the major titratable buffer, but its supply is limited.

A low urine pH alone cannot carry the daily acid load.

Proximal ammoniagenesis

1 glutamine → **Proximal metabolism** → **2 NH_4^+**

Toward urine

2 HCO_3^-

Toward blood

Acidosis increases glutamine metabolism and ammonia production.

Equivalent yield across a multistep pathway, not a single balanced chemical reaction.

Acidosis can increase renal ammonia production substantially.

Ammonia handling across the nephron

Site	Ammonia handling	Direction
Proximal tubule	NH_4^+ production and NHE3 secretion	Cell to lumen
Thick ascending limb	NH_4^+ uses the K^+ site of NKCC2	Lumen → cell → medulla
Collecting duct	NH_3 transfer, including Rhcg	Interstitium / cell to lumen

$$NH_3 + H^+ \rightleftharpoons NH_4^+ \longrightarrow \text{Urine}$$

Acidic lumen favors NH_4^+. Specific transporters can still carry NH_4^+ across membranes.

Final ammonium excretion is an integrated nephron process.

Phosphate and ammonium compared

Feature	Phosphate / titratable acid	Ammonium
Origin	Filtered phosphate buffer	Proximal glutamine metabolism
Urinary product	$H_2PO_4^-$	Mostly NH_4^+
Response to acidosis	Limited by available buffer	Large adaptive increase
New bicarbonate	1 equivalent per net H^+ excreted	1 equivalent per NH_4^+ excreted

Urine pH measures H^+ activity, not the total amount of acid excreted.

Buffer supply and nephron transport determine acid-carrying capacity.

Net acid excretion

Net acid excretion (NAE) =

NH_4^+ + Titratable acid – HCO_3^-

40 + 20 – 5

= 55 mEq/day

All terms are urinary excretion rates.

Use excretion rates, not urine concentrations alone.

Renal responses to acidosis and alkalosis

Variable	Acidosis	Alkalosis, if renal response is intact
HCO_3^- handling	Conserve filtered HCO_3^-	Permit HCO_3^- excretion
H^+ secretion / NAE	Usually increase	Usually decrease
NH_4^+ excretion	Increases with adaptation	Usually decreases
Intercalated-cell role	Alpha: secrete H^+	Beta: apical pendrin secretes HCO_3^-

Pendrin exchanges HCO_3^- into the lumen for Cl^- into the cell.

Volume, chloride, potassium, and filtration can prevent the expected response.

The renal response depends on potassium, chloride, volume, and kidney function.

Factors that modify the renal response

Modifier	Key renal effect	Clinical link
Angiotensin II / low volume	More proximal Na^+ and HCO_3^- reclamation	Maintains alkalosis
Aldosterone	Supports distal H^+ secretion	Deficiency can impair acid excretion
High plasma K^+	Suppresses ammoniagenesis	Type 4 renal tubular acidosis
Reduced functioning nephrons	Lower total acid-excretory capacity	Chronic kidney disease

Tubular defects may impair proximal HCO_3^- reclamation or distal H^+ secretion.

A functioning mechanism can still be limited by nephron number or its environment.

Interpreting an acid–base disturbance

1- Classify pH: acidemia, reference range, or alkalemia.

2- Identify the primary HCO_3^- or PCO_2 change.

3- Calculate expected compensation.

4- Check electrolytes, time course, and clinical context.

Anion gap = Na^+ – (Cl^- + HCO_3^-)

A normal pH can conceal two opposing disturbances.

Expected compensation

Primary process	Expected response, approximate
Metabolic acidosis	$PCO_2 = 1.5 \times [HCO_3^-] + 8 \pm 2$
Metabolic alkalosis	$PCO_2 \approx 40 + 0.7 \times ([HCO_3^-] - 24)$
Respiratory acidosis	HCO_3^- rises ≈1 acute or 3–4 chronic per 10 ↑ PCO_2
Respiratory alkalosis	HCO_3^- falls ≈2 acute or 4–5 chronic per 10 ↓ PCO_2

PCO_2 in mmHg; HCO_3^- in mmol/L. Baseline 40 / 24. Chronic adaptation takes days.

Rules screen for added processes. They are not exact laws or treatment targets.

Compensation rules are approximate physiological checks.

Case 1: Diarrhoea and metabolic acidosis

pH 7.29

HCO_3^- 14 mmol/L

PCO_2 30 mmHg

Acidemia with low bicarbonate: metabolic acidosis.

Anion gap = 140 − (116 + 14) = 10 mmol/L.

Winter’s rule gives 29 ± 2 mmHg. Observed 30 fits.

Intact kidneys should increase NH_4^+ excretion.

Synthetic case. Normal albumin and a usual laboratory anion-gap range assumed.

The respiratory response fits. Healthy kidneys should increase NH_4^+ excretion.

Case 2: Vomiting and metabolic alkalosis

pH 7.50

HCO_3^- 36 mmol/L

PCO_2 48 mmHg

Gastric HCl loss generates metabolic alkalosis.

Expected PCO_2 ≈ 48.4 mmHg. Observed 48 fits.

Low volume promotes proximal bicarbonate reclamation.

Cl^- and K^+ depletion can limit bicarbonate disposal.

Synthetic case: Na^+ 140, Cl^- 90 mmol/L. Clinical context guides interpretation.

Volume, chloride, and potassium depletion can maintain the alkalosis.

Respiratory disorders and renal adaptation

	Chronic respiratory acidosis	Acute respiratory alkalosis
Example	Chronic hypoventilation	Brief hyperventilation
pH / PCO_2 / HCO_3^-	7.35 / 60 / 32	7.49 / 30 / 22
Current HCO_3^- change	Chronic renal adaptation	Mainly early buffering
During renal adaptation	Net acid excretion rises	Net acid excretion falls

PCO_2 in mmHg; HCO_3^- in mmol/L. Synthetic values rounded from the buffer equation.

At the new steady state, net acid excretion again matches the net acid load.

Timing explains why the bicarbonate response differs.

Retrieval questions: renal mechanisms

1 **Reclaiming one filtered HCO_3^- equivalent:**

A Adds one new HCO_3^-
B Preserves the existing bicarbonate pool
C Excretes one NH_4^+
D Requires phosphate loss

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2 **Which alpha-cell apical transporter directly uses ATP?**

A NHE3 B AE1
C H^+-ATPase D NBCe1

Explain the direction and net effect before naming a transporter.

Retrieval questions: balance and compensation

3 **NH_4^+ 40 + titratable acid 20 – HCO_3^- 5 mEq/day**

What is net acid excretion?

A 15 B 55 C 60 D 65 mEq/day

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4 **HCO_3^- 14 mmol/L with PCO_2 40 mmHg:**

A Appropriate respiratory compensation
B Additional respiratory acidosis
C Additional respiratory alkalosis
D Isolated metabolic alkalosis

The bicarbonate loss has a minus sign. Compensation has an expected range.

Integrated renal acid–base balance

Lungs regulate dissolved CO_2

Filtered HCO_3^-	→ **Plasma HCO_3^- pool**	← **Net acid excretion**
Reclamation		**New base supplied**
Existing base conserved	**Nonvolatile acid consumes bicarbonate**	NH_4^+ + titratable acid − urinary HCO_3^-

Arrows into the pool restore bicarbonate. The downward arrow represents bicarbonate use.

Follow both the urinary acid pathway and the blood bicarbonate pathway.

The four primary patterns

Primary process	Initial change	Compensatory direction	Expected kidney response
Metabolic acidosis	↓ HCO_3^-	↓ PCO_2	Conserve HCO_3^-; ↑ acid excretion*
Metabolic alkalosis	↑ HCO_3^-	↑ PCO_2	Excrete HCO_3^- if conditions allow
Respiratory acidosis	↑ PCO_2	↑ HCO_3^-	During adaptation: ↑ acid excretion
Respiratory alkalosis	↓ PCO_2	↓ HCO_3^-	During adaptation: ↓ acid excretion

*Renal failure or tubular dysfunction can prevent the expected response.

Use the pattern table with the calculation, timing, and clinical history.

Five take-home messages

pH reflects the HCO_3^--to-dissolved-CO_2 ratio.

Proximal cells reclaim most filtered bicarbonate.

Reclamation and new bicarbonate generation have different net effects.

NH_4^+ and titratable acid carry almost all urinary acid.

Expected compensation helps reveal a mixed disorder.

Renal acid–base physiology is an exercise in direction, buffering, and balance.

Core reading and supplementary sources

CORE TEXTBOOKS

Tortora GJ, Derrickson B. Principles of Anatomy and Physiology, 16th ed. Ch 26: The Urinary System. Ch 27: Fluid, Electrolyte, and Acid–Base Homeostasis.

Hall JE, Hall ME. Guyton and Hall Textbook of Medical Physiology, 15th ed. Ch 31: Acid–Base Regulation. Elsevier, copyright 2026.

SUPPLEMENTARY PHYSIOLOGY

Hamm et al. Acid-Base Homeostasis. CJASN, 2015.
Weiner & Verlander. Ammonia Transporters and Their Role in Acid-Base Balance. 2017.
Clinical compensation: Merck Manual Professional and American Thoracic Society.

Read the textbook topics alongside the original mechanism diagrams.